

TPA ENGINEERING CORPORATION



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Messrs. HYUNDAI WIA CORPORATION
MEXICO

Date : Nov. 09, 2022

ESTIMATE

No. TPA20221109-02

ENQUIRER : NILL

REF. NO. : NILL

MACHINE : TOYODA CNC Production Anguler - Wheel slide Cylindrical Grinder
model GL5Ai-63

PRICE : F.O.B. Pusan Port in Korea
- details as per attached sheets -

DELIVERY : PLACE : Pusan Port

TIME : 12 months after our receipt of your order note

PAYMENT : T/T (20%:down payment)
(60%:Shipment completed)
(20%: FAC & Warranty Bond Issuance)

VALIDITY : 3 months after the date of this Estimate, or by the date of

We are much pleased in quoting you herein, and thanking for your valued orderin advance.

Yours faithfully,

A handwritten signature in black ink, appearing to be 'JAE HWAN PARK', written over a horizontal line.

JAE HWAN PARK
PRESIDENT

TPA ENGINEERING CORPORATION

[A] TOYODA CNC Angular-Wheelslide Cylindrical Grinder
Model: 60M GL5Ai-63 (wheel dia. of 610 mm Specification)

* Specially designed and tooled up for
finish-grinding of OUTER RACE

		Quantity	Unit	Unit Price	Amount
1	<u>Basic Machine</u>				
1.1	Basic machine Model: 60M GL5Ai-63 Wheel diameter ϕ 610 specification	1	set	168,000,000	168,000,000
2.1	Machine bed	1	set	IIBM	IIBM
2.2	One-body type table	1	set	IIBM	IIBM
2.3	Live spindle workhead	1	set	4,300,000	4,300,000
2.4	Footstock, automatic reposition type, manual taper adjustment type	1	set	7,000,000	7,000,000
2.5	Rotary diamond roll type wheel dresser, mounted on the rear of workhead with the inverter	1	set	7,000,000	7,000,000
2.6	Hydraulic pump unit, common with low oil level warning switch	1	set	IIBM	IIBM
2.7	Lubricating and operating oil pump unit common with low oil level warning switch	1	set	IIBM	IIBM
2.8	Coolant supply unit (1,000L) with magnetic separator, paper filter, oil skimmer, English name plates, Spanish PL name plates, high efficiency motor, coolant pump inverter control, caster and coolant cooler	1	set	27,000,000	27,000,000
2.9	FANUC-make CNC unit	1	set	IIBM	IIBM
2.10	Machine painting (RAL 7035, 1018)	1	set	IIBM	IIBM
2.11	Galvanizing painting	1	set	10,000,000	10,000,000
2.12	Design fee for additional workpieces	1	Lot	9,000,000	9,000,000
2.13	Manual overhead open/shut type full cover and totally-closed cover	1	set	11,000,000	11,000,000
2.14	Gentry loader	1	lot	120,000,000	120,000,000
Sub Total: [1],[2] ₩					363,300,000
3	OPTIONAL ACCESSORIES				
3.1	Automatic electric in-process gauging system, with 1 pce. of "MARPOSS"-make table-mounted type hydraulically-operated gauging head and control unit , pipings, wirings and fittings (HEAD: UNIMAR S10, AMP P7ME), including automatic lateral locator(HEAD:UNIMAR W26, AMP P7ME), mounted on table	1	set	52,500,000	52,500,000
3.2	Master bar (for spindle)	1	set	850,000	850,000
3.3	3-steps signal tower with buzzer	1	set	700,000	700,000
3.4	Cooler for electric control box (RITTAL)	1	set	4,310,000	4,310,000
3.5	Local lighting unit	1	set	770,000	770,000
3.6	Lighting units inside control box	1	set	770,000	770,000
3.7	Electric type jib crane for 0.25ton	1	set	5,900,000	5,900,000
3.8	Air gun & coolant gun	1	set	710,000	710,000
3.9	Mist collector attaching arrangement (excluding mist collector)	1	set	1,030,000	1,030,000
3.10	Terminal box between loader and machine	1	set	900,000	900,000
3.11	Wheel guard auto offset device Front cover automatic correction type, for max. wheel width 125mm	1	set	3,000,000	3,000,000
3.12	Wheel balance unit mounting control	1	set	5,500,000	5,500,000
3.13	Balance arbor	1	set	2,500,000	2,500,000
3.14	Wheel bearing pump unit (separate type)	1	set	2,000,000	2,000,000
Sub Total: [3] ₩					81,440,000

4	<u>Durable tool:</u>			
4.1	Grinding wheel flange (by specification 3sets) (21027-SP301 & 20122-0200 & 21023-0007 commonization)	3 sets	3,500,000	10,500,000
4.2	Feeler & contact for auto-locator (1 set. as spare) (by specification 2sets)	2 sets	1,080,000	2,160,000
4.3	Feeler & contact for auto-sizer (1 set. as spare) by specification 2sets)	2 sets	2,000,000	4,000,000
4.4	Special type work driving center (L-side only) (1set = Installed 1set & Spare 1set) by specification 2sets)	2 sets	3,000,000	6,000,000
4.5	Carbide-tipped work center(R-side only)(1set = Installed 1set & Spare 1set) by specification 2sets)	2 sets	1,500,000	3,000,000
4.6	Driving plate(for attaching drive center) for 3 workpieces (1set = Installed 1set & Spare 1set) (by specification 2sets) (21022-0020&21023-0006 commonization)	4 sets	500,000	2,000,000
4.7	Loader Jaw (1set = Installed 1set & Spare 1set) (21022-0020&21023-0006 commonization) (by specification 2sets)	4 sets	500,000	2,000,000
4.8	Grinding wheel (21027-SP301 2sets / 20122-0200 & 21023-0007 commonization sets)	4 sets	1,800,000	7,200,000
4.9	Diamond roll, (1set = Installed 1set & Spare 1set) (by specification 2sets)	2 sets	2,500,000	5,000,000
4.10	Paper filter (by specification 2sets)	2 sets	1,000,000	2,000,000
			Sub Total: [4]	₩ 43,860,000
5	<u>Supervisor fee:</u>			
5.1	Trial at setup place(MEXICO) (26DAYX 2MAN)	28 M/D	600,000	14,000,000
			Sub Total: [5]	₩ 14,000,000
6	<u>Training & Production support fee:</u>			
6.1	Training(operate 6D X 2M=12M/D & maintenance 6DX2M = 12M/D)	24 M/D	600,000	14,400,000
6.2	Production Support	14 M/D	600,000	7,000,000
6.3	Training translation	10 M/D	400,000	4,000,000
6.4	Manual(usb type 2set, paper type 4set) & Educational Material(paper type 4set)	1 lot	4,000,000	4,000,000
6.5	Air Ticket	5 lot	4,000,000	20,000,000
			Sub Total: [6]	₩ 49,400,000
7	<u>Spare parts</u>			
7.1	Spare parts (3% of Basic machine)			0
			Sub Total: [7]	0
8	<u>Optional items</u>			
8.1	Foundation bar collet type	1	2,500,000	2,500,000
8.2	Work existing confirmation device	1	2,000,000	2,000,000
8.3	Tool kit (with torque wrench)	1	500,000	500,000
9	T/P shipping cost (MEXICO-> TPA)	1	3,000,000	3,000,000
			Sub Total: [8]	₩ 8,000,000
GRAND TOTAL				₩ 560,000,000